SLDDNet: Stagewise Short and Long Distance Dependency Network for Remote Sensing Change Detection

Zhaojin Fu, Jinjiang Li, Lu Ren, and Zheng Chen

***Abstract*— With the rapid development of society, the pace of land change continues to accelerate. Consequently, remote sensing change detection (CD) has become a vital method for monitoring geographical information changes across various domains. However, the increasingly diverse and complex environments and structures where change targets exist pose significant challenges to CD tasks. To address these challenges, we propose a novel approach called stagewise short and long distance dependency network (SLDDNet). SLDDNet uses CNN-Transformer architecture and proposes the Transformer semantic selector to capture long-range feature dependencies and enhance semantic associations from a global perspective. Moreover, the pyramid structure feature stacking is proposed to capture short-range feature dependencies and emphasize local feature information. By integrating these two types of features at each layer, SLDDNet focuses on semantic information and improves its ability to attend to feature details. Furthermore, SLDDNet enhances target position information through axial semantic enhancement and optimizes the network training process using a deep supervision mechanism. Through extensive experiments, SLDDNet outperforms mainstream and state-of-the-art methods on three datasets. Specifically, on the LEVIR-CD dataset, SLDDNet achieves an F1 score of 91.75% and an intersection over union (IoU) of 84.76%. On the WHU-CD dataset, it achieves an F1 score of 92.76% and an IoU of 85.78%. Finally, on the GZ-CD dataset, SLDDNet achieves an F1 score of 86.61% and an IoU of 76.38%.**

***Index Terms*— Attention mechanism, change detection (CD), long and short distance dependency, remote sensing, Transformer.**

I. Introduction

IN THE field of remote sensing, image processing has always been a popular research topic. There is a need for a technology that enables the monitoring of ground change information to facilitate future social development and resource utilization planning. Change detection (CD) tasks precisely fulfill this requirement by detecting differential features captured in different images of the same area at different periods. Through binary maps, CD highlights these differential features in the predicted result image for analysis by researchers. As a downstream task, CD plays a crucial role in various domains, such as urban expansion [1], [2], forest cover [3], [4], land cover [5], [6], [7], [8], military applications [9], and disaster assessment [10], [11].

Over the decades, the development of aerospace technology has led to the launch of satellites [12], [13], [14] by organizations such as the European Space Agency (ESA), the National Aeronautics and Space Administration (NASA) of the United States, and other national and private enterprises, such as Google. These satellites have provided an abundance of satellite and aerial images, which have greatly contributed to the advancement of CD tasks.

However, with the availability of a rich sample space, CD tasks also face significant challenges. Images of the same area captured by satellites and aerial equipment often exhibit variations in color and environment due to seasonal and lighting changes. These variations make the CD tasks potentially interested in environmental changes beyond the changing targets. In addition, the shooting angles of satellites and aerial equipment cause image alignment and lighting problems for targets in the same area, so some of the data need to be image-corrected with the help of detection and enhancement algorithms [15], [16], [17], [18], [19] before they can be applied to the CD tasks.

The field of remote sensing CD tasks has a rich history, with numerous excellent CD algorithms [20], [21], [22] being proposed over the past few decades. In the early stages, CD tasks relied on traditional methods, such as random forests [23], decision trees [24], and support vector machines [25], due to limitations in computer hardware devices such as GPUs. These traditional methods focused on detecting and classifying changed pixels in bitemporal images to generate the final change map. While these methods achieved significant results in some CD tasks, they often lacked advantages in change target recognition and accuracy due to algorithmic threshold limitations and the influence of complex targets and environmental noise present in the images. However, it is crucial to acknowledge the substantial contributions that traditional methods in the field of CD have made to the subsequent advancements in deep learning techniques.

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With the advancements in computer hardware performance and artificial intelligence technology, deep learning has gained widespread applications in the field of CD, giving rise to numerous excellent algorithms [26], [27], [28], [29], [30]. Deep learning, by learning from a sample space, enables the recognition of change targets. Among the various deep learning approaches, the family of algorithms based on convolutional neural networks (CNNs) has gained extensive popularity. CNN-based algorithms outperform traditional methods in handling environmental factors, such as lighting and shadows. Early deep learning strategies are mostly derived from semantic segmentation algorithms and image segmentation algorithms, such as VGG [31], FCN [32], and U-Net [33]. For example, FC-EF [34], based on the U-Net architecture, concatenates bitemporal images and feeds them into the encoder to extract semantic information at different depths. Through the decoder and skip connections, it reconstructs the change targets. While this approach achieves low resource occupancy, the fusion of bitemporal images before feature extraction disrupts the structural information within a single image, leading to suboptimal performance in edge processing for FC-EF.

Subsequently, FC-Siam-Conc [34] employs a dual encoder architecture to process bitemporal images separately. The shallow features captured by the dual encoders are then transmitted to the decoder for fusion in a concatenated manner. This processing approach has become one of the mainstream ideas in the field of CD. Conversely, FC-Siam-Di [34] takes an opposite approach during the fusion of shallow features. It subtracts the two types of feature information and computes the absolute value before feeding them into the decoder for change target reconstruction. This processing approach has also emerged as a mainstream idea in subsequent developments.

The progress in ideas and technologies within the field of CD has revealed the limitations of traditional CNN-based methods. Traditional CNNs, constrained by convolutional kernels, are unable to consider long-range dependencies among features and can only establish relationships within the kernel's feature information. To address this issue in CD tasks involving heterogeneous remote sensing images, Lv et al. [35] propose combining multiple-scale convolutional modules with the U-Net architecture to expand the model's field of view. This expansion enables the model to capture change targets of various sizes. In DSA-Net [36], ASPP [37] is introduced to capture features from even longer distances. While this approach expands the network's field of view and obtains multiscale feature information to some extent, it still falls short of establishing a global perspective. DSA-Net compensates for deep semantic information using ASPP, but it fails to account for the loss of detailed information in shallow networks due to downsampling. Consequently, the prediction results of DSA-Net exhibit suboptimal performance in capturing fine-grained details of change targets.

Although expanding the field of view has improved the accuracy and performance of CD tasks to a certain extent, it still does not effectively capture global features. To overcome this limitation, CNN architectures based on attention mechanisms are gradually being applied in CD tasks. MACGGNet [38] proposes the position-channel attention module to enhance the network's focus on spectral and spatial information. In addition, MACGGNet uses multiscale attention modules to capture richer feature information from bitemporal images. IFNet [39] incorporates channel attention [40] to enhance the relevance of pixels in the fused features and employs spatial attention [41] to strengthen the position information of differentiating targets. Similarly, in DTCDSCN [42], channel attention and spatial attention are utilized for feature enhancement as well. However, the difference lies in the fact that DTCDSCN combines the two attentions in a parallel manner. AMIO-Net [43] proposes the pyramid pooling attention module in the encoder stage to capture more diverse and global information through parallel multiscale pooling, thus enhancing the strength of high-level semantic information. In the decoder stage, AMIO-Net incorporates the siamese attention mechanism module to restore fine-grained details from shallow features.

The self-attention mechanism proposed in Transformer [44] has proven effective in capturing long-range dependencies among feature information, making it widely used in the field of remote sensing [45], [46], [47], [48] for its ability to acquire global feature information. BIT [45] utilizes a Transformer encoder to establish global feature relationships for fused high-dimensional information, thereby enhancing the strength of semantic information. However, BIT's use of ResNet_{18} [49] for feature information at different scales has limitations, and it lacks finesse during image restoration and difference labeling in the decoding stage. Change former [46], which appeared later, employs multihead self-attention modules as the backbone network for feature information extraction, but it also significantly increases resource utilization. ICIF-Net [47] adopts a parallel approach, using both CNN and Transformer backbone networks to extract feature information, and creates a dual-branch multilevel intertemporal network to fuse the feature information obtained by the two different backbone networks. ICIF-Net's approach has yielded good results, extracting feature information from both local (CNN backbone) and global (Transformer backbone) perspectives. However, there is no interaction between the two feature information extraction processes in ICIF-Net.

To address the challenges faced by CD tasks and the limitations of existing methods and architectures, we propose a stagewise short and long distance dependency network (SLDDNet), as shown in Fig. 1. Specifically, we utilize the idea of residual structure to propose pyramid structure feature stacking (PSFS), which establishes short-range dependencies for local feature information in a central aggregation manner. This enhances the network's ability to capture local features, enabling it to effectively address small changes in targets. In addition, we propose Transformer semantic selector (TSS), which utilizes self-attention to establish long-range dependencies for feature information from a global perspective. This enhances the network's ability to capture global features, allowing it to maintain feature continuity when facing large changes in targets. At each layer, SLDDNet employs a stagewise approach to fuse global and local features and feed them into the next layer. The stagewise processing approach,

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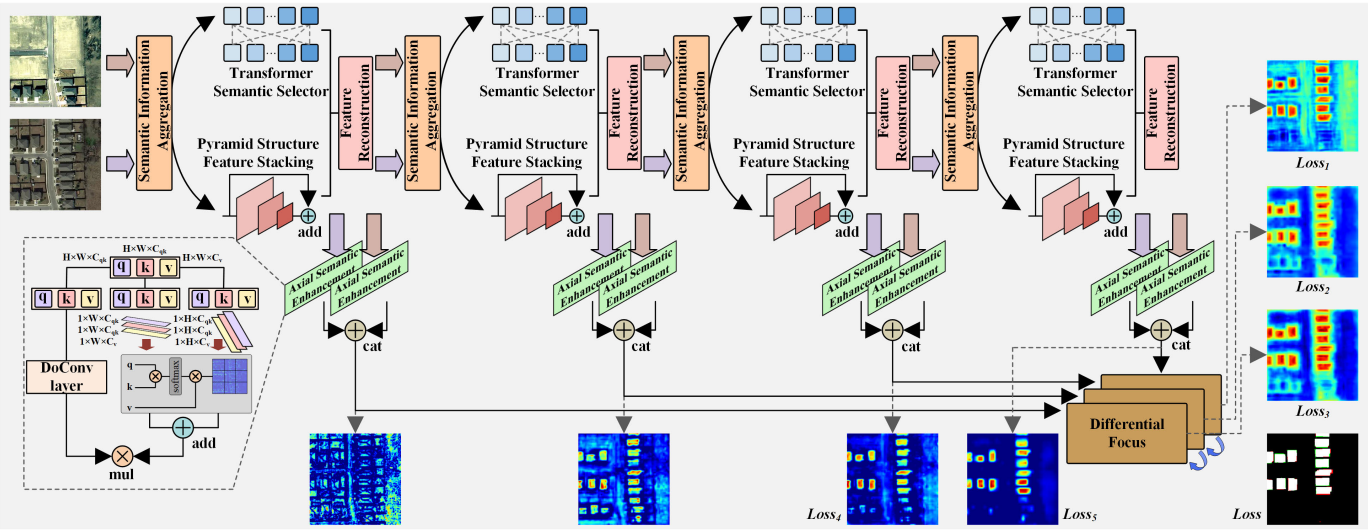


Fig. 1. SLDDNet. TSS is used to obtain long-range dependencies of features. PSFS is used to obtain short-range dependencies of features. ASE is used to enhance the position information of features. DF is used to accomplish the capture of differential features. The network uses a deeply supervised mechanism to calculate losses on the multiscale output results.



Fig. 2. Presentation of SLDDNet results on three datasets. Red represents misidentified regions, and green represents unidentified regions.

while maximizing the preservation of feature information, allows PSFS and TSS to share the feature information that they individually fail to capture. As a result, this enables the interaction and compensation between global and local feature information, providing a rich semantic context for extracting differential features. More importantly, this processing approach optimizes the feature interaction issue that CNN and Transformer architectures face during feature extraction in the current state-of-the-art methods in the field of CD. Furthermore, axial semantic enhancement (ASE) [50] is employed to enhance feature information at each stage. Finally, we propose differential focus (DF) for significant labeling of differential features.

Fig. 2 presents a comparison of the results between SLDDNet and mainstream CNN-Transformer hybrid architecture CD methods on three datasets. The visualization overlays the results, where red indicates misidentified areas and green represents undetected areas. BIT utilizes ResNet_{18} for feature extraction from bitemporal images, while change former employs Transformer as the backbone network. However, neither a single CNN nor a Transformer architecture for feature extraction demonstrates significant advantages in predicting change targets. ICIF-Net combines independent CNN and Transformer for feature extraction, but it fails to fuse the local features obtained by CNN and the global features obtained by Transformer at the same scale, leading to errors in edge recognition of change targets. In contrast, SLDDNet integrates long and short distance dependencies during feature extraction at the same scale, resulting in richer bitemporal features, improved edge recognition, and better feature completeness compared to current mainstream methods.

Overall, our contribution to this work has been mainly as follows.

1) SLDDNet proposes a new CNN-Transformer hybrid encoder for accomplishing feature extraction of bitemporal images. By parallelizing PSFS with TSS, it solves the existing methods that only use a single CNN or Transformer architecture for feature extraction. The proposed staged long–short distance-dependent aggregation optimizes the feature interaction problem that exists when existing methods employ a dual backbone network of CNN and Transformer for feature extraction.
2) DF is proposed to compensate for the detail of the semantic information obtained by the deeper network using the semantic information obtained by the shallow network, significantly labeling the targets of change.
3) The ASE is introduced in the network to enhance the positions of the feature information obtained at each stage. A deep supervision mechanism is adopted to optimize the network training process and enhance supervision over the hidden layers. DO-Conv is used to replace traditional convolution to obtain richer local detail information.

II. Related Work

This section will briefly describe the CNN-Transformer-based remote sensing CD method, followed by an introduction to Transformer applied in image processing tasks, and finally an introduction to DO-Conv.

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A. Remote Sensing Change Detection Based on CNN-Transformer

Many of the current state-of-the-art CD methods employ a hybrid CNN-Transformer architecture. The self-attention mechanism proposed by Transformer allows the model to capture the relationships between each pixel and all other pixels in the image, enabling the modeling of long-range dependencies. On the other hand, CNNs excel at acquiring feature information in different dimensions with fewer parameters and possess strong local perception capabilities. To better monitor changes in farmland areas, MSCANet [51] adopts a CNN-Transformer architecture. It first extracts feature information of different dimensions using ResNet and then creates a token encoder with spatial attention and multihead self-attention to enhance the global feature information. The use of spatial attention helps reduce the computational cost of self-attention by changing the dimensionality of the feature information.

DBFGAN [52] introduces a feature-guided aggregation module in the encoder stage, employing a dual-branch processing approach with CNN and Transformer. The two branches independently process feature information: in odd-numbered layers, CNN captures features that are then passed to Transformer for feature compensation, while, in even-numbered layers, Transformer captures features that are then passed to CNN for feature fusion. This approach combines positional information with global semantic information, enhancing the network's ability to learn and recognize differential features. PyramidSCDFormer [53] utilizes self-attention to create a shunted self-attention (SSA), which extracts multiscale features. In the encoder stage, PyramidSCDFormer uses SSA to enhance the feature information captured by each layer of the pyramid structure, leading to promising results.

FTN [54] adopts a CNN-Transformer architecture to effectively capture change targets. To enhance feature extraction from bitemporal images with richer semantic information, FTN incorporates the Swin Transformer [55] as its backbone network. The multiscale features obtained from the FTN backbone network are then processed through a pyramid structure and the progressive attention module, facilitating the effective aggregation of different semantic information. This integration enables the fusion of deep semantic information with detailed shallow features, ultimately enabling the detection of change targets. However, FTN lacks sufficient attention to local semantic information during the feature extraction stage, resulting in limited performance in capturing edge features.

In the field of CD, many excellent algorithms are achieving impressive results. Our research indicates that most CD architectures currently combine CNN and Transformer. However, in terms of feature extraction, single CNN or Transformer architectures are still commonly employed. Although a few networks integrate CNN and Transformer in the backbone network for feature extraction, the information interaction during the extraction process is not comprehensive. Therefore, we propose the creation of SLDDNet, a network that can effectively stage the mutual compensation of global semantic information and local detailed information.

B. Transformer in Image Processing

In the past, CNN has been the predominant method in the field of image processing, with most network architectures incorporating various forms of attention mechanisms based on CNN. However, the emergence of vision transformer (ViT) [56] disrupted the dominance of CNN in image processing. ViT divides the image into patches of equal size and utilizes self-attention to capture global semantic information. Self-attention involves matrix multiplication, leading to exponential computational complexity with larger input images. ViT's image block processing significantly reduces computational complexity and provides important insights for the advancement of Transformers in image processing.

The Swin Transformer is a novel architecture that builds upon ViT by incorporating multiscale processing through varying image sizes and the number of patches at each layer. It also introduces sliding window computation, limiting attention calculation within a window to further reduce computational cost. Similarly, PVT [57] reduces image resolution in each layer's patch embedding to achieve multiscale processing while modifying the sizes of key K and value V in multihead self-attention to decrease computational requirements. Recognizing the memory bandwidth bottleneck, Rev-MViT [58] introduces a reversible ViT that decouples GPU memory requirements from changes in model depth, resulting in significant memory usage reduction. Rev-MViT establishes a residual path for self-attention to effectively mitigate gradient vanishing in deep networks.

Based on these remarkable architectures, it is evident that most self-attention processes rely on matrix operations and rarely employ other processing methods. Therefore, combining convolution operations with self-attention is expected to enhance the selection of semantic intensity. To address this, the TSS is being developed in each stage of the SLDDNet backbone network.

C. DO-Conv

The DO-Conv [59] combines the calculation of traditional convolution with depthwise convolution [60]. Specifically, the region covered by the convolution kernel can be represented as $\mathbb{P} \in R^{H\times W\times C_{\text{in}}}$, where C_{in} is the number of input channels. The convolution kernel of depthwise convolution can be represented as $\mathbb{D} \in R^{H\times W\times D_{\text{mul}}\times C_{\text{in}}}$, where D_{mul} is the depth multiplier. In DO-Conv, the convolution kernel of traditional convolution can be represented as $\mathbb{W} \in R^{C_{\text{out}}\times D_{\text{mul}}\times C_{\text{in}}}$, where C_{out} is the output channel. The DO-Conv convolution operation can be represented as $\circledast$ as follows:

$$\mathbb{O} = (\mathbb{D}, \mathbb{W}) \circledast \mathbb{P}. \tag{1}$$

During the calculation process, $\mathbb{D}$ is transposed to obtain $D^T \in R^{D_{\text{mul}}\times(H\times W)\times C_{\text{in}}}$, and $\mathbb{W}'$ is obtained by calculating D^T with $\mathbb{W}$, as in (2). Then, $\mathbb{W}'$ is used to calculate with $\mathbb{P}$ to obtain the final result, as in (3)

$$\mathbb{W}' = D^T \circ \mathbb{W} \tag{2}$$

$$\mathbb{O} = \mathbb{W}' * \mathbb{P}. \tag{3}$$

Two convolution operators' precomputation reduces the overall computational cost of DO-Conv. As DO-Conv embeds

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Algorithm 1 SLDDNet

Input: $\mathrm{Im}age_1 = x_i \ i \in [1, n]$
$\mathrm{Im}age_2 = y_i \ i \in [1, n]$
$GT = G_i \ i \in [1, n]$
while *not converge* **do**
Push(zip(x_i, y_i, G_i)) in
$list_x = [], list_y = []$
for j to 4 **do**
$x_i = SIA_j(x_i), y_i = SIA_j(y_i)$
$x_{tss} = TSS_j(x_i), x_{psfs} = PSFS_j(x_i)$
$y_{tss} = TSS_j(y_i), y_{psfs} = PSFS_j(y_i)$
$list_x.append\left(FR_j\left(x_{tss} + x_{psfs}\right)\right)$
$list_y.append\left(FR_j\left(y_{tss} + y_{psfs}\right)\right)$
end for
$list_out = []$
for j to 4 **do**
$x_{ase} = ASE_j(list_x[j])$
$y_{ase} = ASE_j(list_y[j])$
$list_out.append(f_{1\times1}(cat(x_{ase}, y_{ase})))$
end for
$Out = DF(list_out[3], list_out[2])$
$Out = DF(list_out[1], Out)$
$Out = DF(list_out[0], Out)$
$Out = sigmoid(Out)$
end while

depthwise convolution into traditional convolution computation, it can achieve better training results during the training process. In SLDDNet, we use DO-Conv to replace all traditional convolutions, which helps to improve the training performance of the network.

III. Method

In this section, we will provide a detailed introduction to the main structure of the network, including the TSS, DF, and the introduced ASE. Finally, we will demonstrate the algorithmic flow of SLDDNet, as in Algorithm 1.

A. Network Structure

Fig. 2 illustrates the overall architecture of SLDDNet, where we adopt a stagewise semantic reconstruction approach to extract feature information. Each stage consists of four parts: semantic information aggregation (SIA), TSS, PSFS, and feature reconstruction (FR). In SIA, we use a semantic aggregation operator $Q \in \mathbb{R}^{3\times3\times C}$ with an aggregation range of 3×3. The semantic aggregation operator Q recalculates the weights of feature $O \in \mathbb{R}^{H\times W\times C}$ in a traversal form and updates it to obtain a new aggregated semantic O', as follows:

$$O' = \sum_{i=1}^{H-1}\sum_{j=1}^{W-1} Q \circledast x_{(i,j)} \tag{4}$$

where $x_{(i,j)} \in O$.

Next, the CNN-Transformer parallel architecture is employed to capture both short and long distance dependencies. The PSFS is utilized to establish short-range dependencies among feature information through central aggregation,

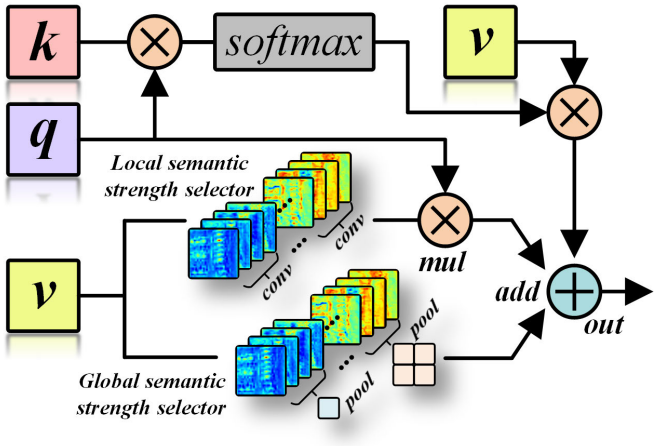


Fig. 3. TSS.

thus reinforcing the network's attention to local features. PSFS consists of three layers, where each layer incorporates convolutional kernels of different sizes to capture short-range dependencies at various distances. This approach enhances the network's capability to focus on targets with small variations. In addition, PSFS addresses missing feature information through residual connections suppresses irrelevant channels using channel attention and ultimately produces the desired results, denoted as P_{out}, as in (5) and (6). Here, $x \in \mathbb{R}^{H\times W\times C}$, $f_{\text{PSFS}}(\cdot)$ represents PSFS, GAP represents global average pooling, W_1 and W_2 represent fully connected layers, σ represents ReLU, and δ represents sigmoid

$$\text{out} = f_{\text{PSFS}}(x) + x \tag{5}$$

$$P_{\text{out}} = \sigma(W_2\delta(W_1\text{GAP}(\text{out}))). \tag{6}$$

The corresponding counterpart to PSFS is the TSS, as illustrated in Fig. 3. TSS is designed to establish long-range dependencies among feature information. We have extended the self-attention by proposing the local semantic strength selector and the global semantic strength selector for the processing of value V. This extension allows for the capture of detailed information and target position information, expanding the network's attention to semantic information without compromising long distance dependency relationships. This approach ensures that the network maintains strong feature continuity when dealing with large-scale change targets.

In many architectures that combine CNN and Transformer in parallel, there is often a lack of interaction between the feature information captured by the two architectures. This means that each architecture's source of information remains limited to itself. To address this limitation, we introduce the FR module at the end of the staged processing. FR does not employ a complex structure but simply adds the two types of feature information and applies it to smooth through a convolutional layer. The fused information is then passed on to the next stage for further processing. It is worth noting that, in the SIA, downsampling is applied to enable SLDDNet to capture multiscale feature information.

In CD tasks, the presence of large shadows around buildings due to lighting conditions often hinders the accurate

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reconstruction of change features, resulting in shifted results. To mitigate this issue, we introduce the ASE. It compresses the different scale feature information captured by the backbone network horizontally and vertically, and feeds it into self-attention to enhance the position information, thereby improving the network's ability to handle challenging lighting conditions and produce more precise results.

Finally, the bitemporal features processed by ASE are concatenated and sent to DF for differential feature attention through 1×1 convolutional operation. As in Fig. 2 (Loss_i), we adopt a deep supervision mechanism to supervise the fusion of deep semantic information and the DF, and feedback to the network training process through loss computation. Specifically, each supervised layer outputs a result that is convolution and sigmoid processed, and fed into a binary cross-entropy loss with size-matched ground truth (GT) to calculate the loss. Finally, Loss_i is superimposed on the loss calculated from the final SLDDNet result as the final loss function.

B. Transformer Semantic Selector

In images, regions with large color differences and significant gray-scale variations contain obvious edge and detail information, and these regions have high semantic information intensity. On the other hand, regions with smooth colors and gradual gray-scale changes contain macroscopic positional information about the target, and these regions have low semantic information intensity. Although self-attention can establish long-range dependencies between feature information, there is still room for improvement in terms of semantic information acquisition. Therefore, we propose TSS (as in Fig. 3).

TSS still uses the matrix calculation method of QKV and employs multihead self-attention to ensure that remote dependency relationships can be established among feature information, as in (7)–(9). The final result is Out_{att}, where $W_j^Q \in \mathbb{R}^{C_i \times d_{\text{head}}}$, $W_j^K \in \mathbb{R}^{C_i \times d_{\text{head}}}$, $W_j^V \in \mathbb{R}^{C_i \times d_{\text{head}}}$, and $W^O \in \mathbb{R}^{C_i \times C_i}$ are linear projection parameters, h is the number of attention heads, $\text{Cat}(\cdot)$ represents the concatenation operation, $\text{Attention}(\cdot)$ represents the self-attention calculation, and $\text{soft}\max(\cdot)$ is the normalization function

$$\text{Out}_{\text{att}}(Q, K, V) = \text{Cat}(\text{head}_0, \ldots, \text{head}_h) W^O \tag{7}$$

$$\text{head}_j = \text{Attention}\left(QW_j^Q, KW_j^K, VW_j^V\right) \tag{8}$$

$$\text{Attention}(q, k, v) = \text{soft}\max\left(\frac{qk^T}{\sqrt{d_{\text{head}}}}\right) v. \tag{9}$$

For value V, in addition to being involved in matrix operations, it is also input into the local semantic strength selector and the global semantic strength selector to obtain detailed and positional information, enabling the network to maintain good feature continuity when recognizing large-scale change targets. Specifically, in the local semantic strength selector, we divide value V into three groups according to channels, obtain v_i, calculate feature weights w_i using convolution kernels of 3×3, 5×5, and 7×7, respectively, and finally reweight them into v_i and recombine them through channel concatenation to obtain the result V_L, as in (10).

Capturing detail information will also strengthen the noise in the image. Therefore, we multiply query Q with V_L to fill in possible noise with original features, as in (11). Here, $\odot$ is the element-by-element multiplication, and n is the number of groups. In the global semantic strength selector, we also divide the value V into groups to obtain v_j. However, in order to capture location information from a macro perspective, we use average pooling with different pooling kernels to calculate weights p_j for the grouped features. We assign the weights to the features and recombine them, as in (12), where m is the number of groups. Finally, the feature enhanced by long-range dependency of multihead self-attention (Out_{att}) and the feature enhanced by local semantic strength (V_L') and global semantic strength (V_G) are added together to obtain the final output result Out, as in (13)

$$V_L = \text{Cat}(v_i w_i, \ldots, v_n w_n) \tag{10}$$

$$V_L' = Q \odot V_L \tag{11}$$

$$V_G = \text{Cat}\left(v_j p_j, \ldots, v_m p_m\right) \tag{12}$$

$$\text{Out} = \text{Out}_{\text{att}} + V_L' + V_G. \tag{13}$$

C. Axial Semantic Enhancement

To enhance the position information, we introduce ASE into the network, as in Fig. 2. The features are first expanded by 1×1 convolution to obtain $Q \in \mathbb{R}^{H \times W \times C_{qk}}$, $K \in \mathbb{R}^{H \times W \times C_{qk}}$, and $V \in \mathbb{R}^{H \times W \times C_v}$. Q, K, and V are compressed along the horizontal direction to obtain $q_{(h)}, k_{(h)} \in \mathbb{R}^{H \times C_{qk}}$ and $v_{(h)} \in \mathbb{R}^{H \times C_v}$, and compressed along the vertical direction to obtain $q_{(v)}, k_{(v)} \in \mathbb{R}^{W \times C_{qk}}$ and $v_{(v)} \in \mathbb{R}^{W \times C_v}$. To enable multihead self-attention to focus on position information, ASE uses $q_{(h)}$ and $k_{(h)}$ to generate learnable position embedding parameter $r_{(h)}^q, r_{(h)}^k \in \mathbb{R}^{H \times C_{qk}}$ through linear interpolation and uses $q_{(v)}$ and $k_{(v)}$ to generate $r_{(v)}^q, r_{(v)}^k \in \mathbb{R}^{W \times C_{qk}}$ in the same way. ASE takes the horizontal and vertical feature components and sends them separately into the multihead self-attention to focus on both global semantic information and positional information, as in (14), where $\text{sm}(.)$ is softmax. Finally, to compensate for the lost detail features during compression, ASE combines the original features and the global semantic features captured by multihead self-attention through matrix multiplication to obtain the final output

$$\begin{aligned}\text{Att}_{(i,j)} = &\sum_{p=1}^{H} \text{sm}_p\left(\left(q_{(h)i} + r_{(h)i}^q\right)^\top \left(k_{(h)p} + r_{(h)p}^k\right)\right) v_{(h)p} \\ &+ \sum_{p=1}^{W} \text{sm}_p\left(\left(q_{(v)j} + r_{(v)j}^q\right)^\top \left(k_{(v)p} + r_{(v)p}^k\right)\right) v_{(v)p}.\end{aligned} \tag{14}$$

D. Differential Focus

Considering the fact that the reconstruction of the change targets requires both deep semantic information and a large amount of detailed information captured by shallow networks, we propose DF as in Fig. 4.

In DF, the shallow semantic information (X_l) and deep semantic information (X_d) are first position-encoded.

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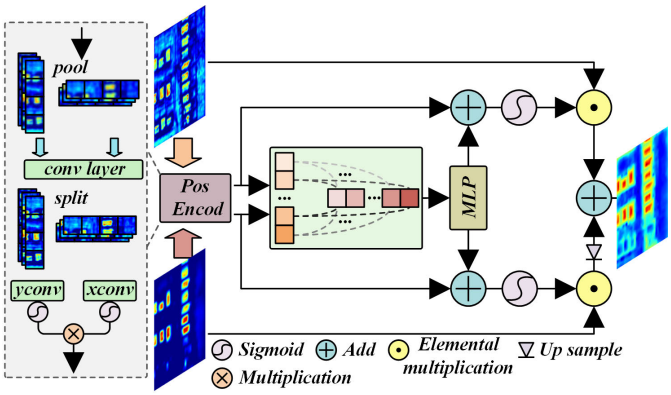


Fig. 4. DF block.

Specifically, the input features are processed by horizontal pooling and vertical pooling to form a set of position-aware features, as in (15) and (16). Then, the position-aware features are recombined by concatenation and channel integration using a 1×1 convolution, resulting in feature (P) as in (17). After decomposition, the horizontal vector (P^w) and the vertical vector (P^h) are processed by horizontal convolution ($f^w_{1\times1}(\cdot)$) and vertical convolution ($f^h_{1\times1}(\cdot)$), respectively, resulting in two different sets of feature weights that are weighted and added to the original features to obtain X'_l and X'_d, as in (18)

$$z^h(h) = \frac{1}{W}\sum_{0\le i\le W} x(h,i) \tag{15}$$

$$z^w(w) = \frac{1}{H}\sum_{0\le j\le H} x(j,w) \tag{16}$$

$$P = \delta f_{1\times1}\left(\mathrm{Cat}\left(z^h, z^w\right)\right) \tag{17}$$

$$X' = X \times \sigma\left(f^h_{1\times1}\left(P^h\right)\right) \times \sigma\left(f^w_{1\times1}(P^w)\right). \tag{18}$$

To capture differential features, X'_l and X'_d are decomposed into k groups of length r, denoted as ($X''_l, X''_d \in \mathbb{R}^{k\times r}$) in (19). The differential matrix S is calculated through matrix computation. The weights of X'_l and X'_d are updated by S, as in (20). Finally, the differential features' position and detail information are reconstructed by elementwise multiplication with the original features, resulting in the final prediction result O_{DF}, as in (21)

$$S = X''_l X''^{\top}_d \tag{19}$$

$$X' = \sigma\left(X' + fc(\mathrm{flatten}(S))\right) \tag{20}$$

$$O_{\mathrm{DF}} = X_l \odot X'_l + \left(X_d \odot X'_d\right) \uparrow . \tag{21}$$

IV. Experiment

In this section, we demonstrate the experimental process. First, we present the relevant experimental parameters, datasets involved, and comparison methods. Then, we show the ablation experiments conducted during the model determination process. Finally, we present the details of the comparative experiments.

A. Experimental Parameter

We use Linux as the system environment, Python version 3.8, and PyTorch version 1.4.0. All comparison methods are powered by NVIDIA TITAN RTX.

The highest number of iterations for all comparison methods involved in the experiment is 200 epochs. After each epoch of training, the current optimal model is validated and saved. In terms of the learning rate, we use the commonly used value of 0.001 in the CD field and use Adam as the optimizer. All methods use binary cross-entropy loss as the loss function, as in (22), where $p(\cdot)$ is the GT pixel value and $\hat{p}(\cdot)$ is the resulting pixel value. Due to the adoption of the deep supervision mechanism, the network calculates additional losses for five different scales of accompanying outputs. Based on previous work experience, the weights of the accompanying losses are set to 0.5 and 0.2, respectively. Therefore, the total loss L is given in (23)

$$L_{\mathrm{CE}(p,\hat{p})} = -\frac{1}{WH}\sum_{x=0}^{W-1}\sum_{y=0}^{H-1} p(x,y)\log\hat{p}(x,y) + (1-p(x,y))\log\left(1-\hat{p}(x,y)\right) \tag{22}$$

$$L = 0.5(\mathrm{Loss}_1 + \mathrm{Loss}_2 + \mathrm{Loss}_3) + 0.2(\mathrm{Loss}_4 + \mathrm{Loss}_5) + \mathrm{Loss}. \tag{23}$$

We evaluate the performance of each method using seven evaluation metrics and calculated them using PyCharm. The seven evaluation metrics are overall accuracy (OA) [as in (24)], average accuracy (AA) [as in (25)], kappa [as in (26)], F1 [as in (27)], intersection over union (IoU) [as in (28)], precision [as in (29)], and recall [as in (30)].

Here, TP represents true positive, TN represents true negative, FP represents false positive, and FN represents false negative. N represents the number of classes, which is 2 in binary classification tasks. In binary classification tasks, OA is used to calculate the ratio of correctly classified samples to the total number of samples. Its maximum value is 1. The closer the model's result is to 1, the better its performance. In classification tasks, AA assigns equal weights to different classes and calculates the AA across different classes. Its maximum value is 1.

Kappa is used to measure the consistency between the model's predictions and the GT. It considers the issue of random classification in the calculation process, which allows for a more accurate evaluation of model performance. The higher the model's performance, the closer the kappa value is to 1. In (26), Po represents the actual accuracy achieved by the model on the dataset, while Pe represents the accuracy expected by chance.

F1 score combines both precision and recall with equal weights. Its maximum value is 1, and it effectively reflects the model's accuracy and its ability to correctly identify positive samples. IoU represents the overlap between the model's predicted results and the GT, with a maximum value of 1. Precision represents the proportion of true positive samples among the samples predicted as positive, while recall represents the ratio of correctly identified positive samples to the actual number of positive samples. The maximum value for both precision and recall is 1

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \tag{24}$$

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$$\mathrm{AA} = \frac{1}{N}\sum_{N}^{i=1}\frac{\mathrm{TP}_i + \mathrm{TN}_i}{\mathrm{TP}_i + \mathrm{TN}_i + \mathrm{FN}_i + \mathrm{FP}_i} \tag{25}$$

$$\mathrm{Kappa} = \frac{\mathrm{Po} - \mathrm{Pe}}{1 - \mathrm{Pe}} \tag{26}$$

$$\mathrm{F1} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{27}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{FP} + \mathrm{FN} + \mathrm{TP}} \tag{28}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{29}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{30}$$

B. Experimental Dataset

LEVIR-CD [61] is an optical dataset widely used in the remote sensing CD field in recent years, collected from Google Earth. The dataset consists of 637 very-high-resolution (VHR, 0.5 m/pixel) images, each with a size of 1024 × 1024. It covers different districts in several cities in Texas, including residential areas, commercial areas, and industrial areas. The bitemporal images in this dataset record a large number of changes in regions and buildings. Due to the different times of image acquisition, seasonal and lighting changes have some impact on the samples, which poses a challenge to the performance of the network. The default crop size of LECIR-CD is 256 × 256, and its training set, validation set, and test set are randomly sampled according to the default division ratio (7:1:2), resulting in sizes of 7120, 1024, and 2048, respectively.

WHU-CD [62] is an optical dataset collected from aerial devices, which covers a large amount of building information in Christchurch, New Zealand. The resolution of VHR images is 0.075 m/pixel. Due to its high resolution, the details of the buildings are clearer, which makes the shadows around the buildings caused by lighting more obvious. During experiments with this dataset, models often experience target position deviation due to these environmental factors. The original size of the aerial image is 32 507 × 15 354, which cannot meet the memory requirements. We also perform nonoverlapping cropping of this dataset to 256 × 256, and since the amount of data in WHU-CD is relatively small, we divide it according to the ratio (8:1:1) to get the training set, the validation set, and the test set as 6096, 762, and 762, respectively.

GZ-CD [63] is collected from Google Earth and covers the changes in Guangzhou, China, between 2006 and 2019. The GZ-CD dataset is an optical dataset that includes VHR images with original sizes ranging from 1006 × 1186 to 4936 × 5224 and a resolution of 0.55 m/pixel. The dataset also includes different types of buildings, such as residential and industrial areas, which have different colors, adding difficulty to the recognition of the changing targets. The GZ-CD is cropped to a size of 256 × 256 without overlap and divided according to the ratio (8:1:1) to obtain the training, validation, and test sets as follows: 2834, 400, and 325, respectively.

C. Comparison Method

To comprehensively compare the performance of the network, we select mainstream and state-of-the-art methods for comparison. This includes CNN-based methods, FC-Siam-Conc; three attention-based methods, IFNet, SNUNet, and DTCDSCN, and five CNN-Transformer-based methods, MSCANet, BIT, change former, FTN, and ICIF-Net. Next, we will briefly describe the main ideas of these nine methods.

FC-Siam-Conc [34] adopts the U-Net architecture. The bitemporal features transmitted by the shallow network are fused through concatenation and are integrated with deep semantic information.

IFNet [39] adopts channel attention and spatial attention to enhance the extraction of difference features. The deep supervision mechanism is used to supervise the process of difference feature extraction.

SNUNet [64] uses the UNet++ architecture. It compensates for details and semantic information through dense skip connections. An ensemble channel attention module is created to process multiscale semantic information.

DTCDSCN [42] utilizes SE-ResNet as the backbone network and creates a dual attention module with channel attention and spatial attention to enhance feature information.

MSCANet [51] utilizes ResNet as the backbone network for feature extraction of spatiotemporal images. It uses spatial attention and Transformer to create a token encoder that enhances global semantic information.

BIT [45] uses ResNet as the backbone network. Transformer is used to establish long-range dependencies for deep semantic information. Detail semantic information is emphasized by subtracting and taking the absolute value of the difference features.

Change former [46] uses the Transformer architecture as the backbone network. It creates a difference module to continuously aggregate the difference features in each layer of the feature information.

FTN [54] uses Swin Transformer to complete feature extraction and obtain multiscale feature information. FTN combines a pyramid structure with a progressive attention module for receiving the multiscale information obtained by the backbone network and completes the attention on the changing targets. Through deep supervised learning, FTN calculates losses for each layer of the pyramid structure's output.

ICIF-Net [47] utilizes a dual backbone network consisting of a CNN and a Transformer to extract features, but the feature extraction processes of the two backbone networks have no interaction. To address this issue, Conv attention is created to fuse features of the same resolution.

D. Ablation Experiment

During the process of determining the final network architecture, we conduct repeated experiments on the effectiveness of different modules. We select the five most representative ablation models for a detailed introduction and show the experimental process on three datasets. All experimental results are visually covered with GT, where red represents the misidentified area and green represents the unidentified area.

1) Model Introduction: The specific modules used for the five ablation models are illustrated in Table I.

In order to verify the influence of the local short distance dependency relationship obtained by PSFS on the overall architecture of the network, we disable the use of PSFS in

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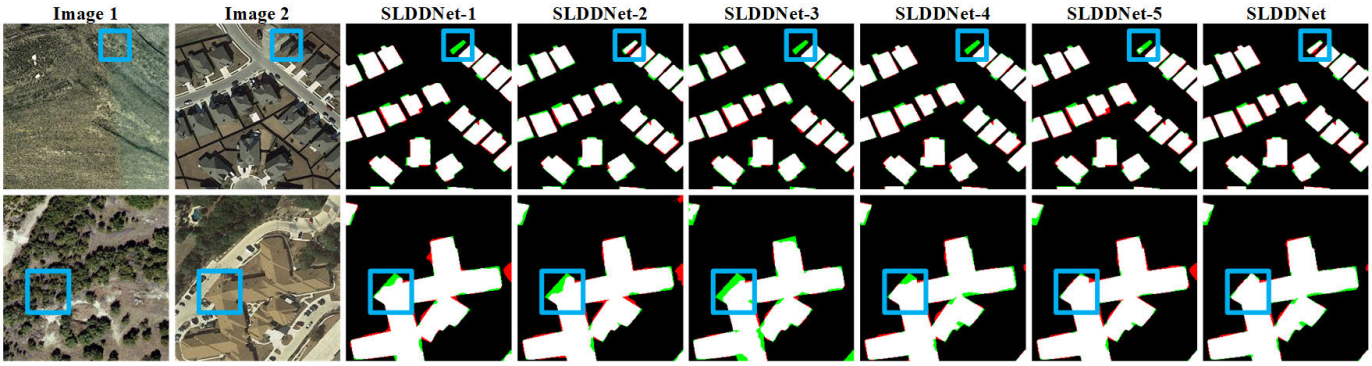


Fig. 5. Results of ablation experiments on LEVIR-CD.

TABLE I
ABLATION MODEL DEMONSTRATION

Method	PSFS	TSS	ASE	Deep supervision	DF
SLDDNet-1	×	✓	✓	✓	✓
SLDDNet-2	✓	×	✓	✓	✓
SLDDNet-3	✓	✓	×	✓	✓
SLDDNet-4	✓	✓	✓	×	✓
SLDDNet-5	✓	✓	✓	✓	×
SLDDNet	✓	✓	✓	✓	✓

TABLE II
INDICATORS FOR FIVE ABLATION MODELS ON THE LEVIR-CD DATASET

Method	OA	AA	Kappa	F1	IoU	Precision	Recall
SLDDNet-1	99.12	99.12	90.84	91.31	84.01	91.35	91.27
SLDDNet-2	99.11	99.11	90.74	91.20	83.83	92.21	90.22
SLDDNet-3	99.13	99.13	88.68	91.32	84.02	92.64	90.03
SLDDNet-4	99.14	99.14	91.08	91.54	84.39	92.12	90.96
SLDDNet-5	99.15	99.13	90.87	91.58	84.46	92.18	90.98
SLDDNet	99.18	99.16	91.27	91.75	84.76	93.64	89.94

SLDDNet-1. The phased processing method loses its effectiveness in the single Transformer architecture of SLDDNet-1.

Similarly, in SLDDNet-2, to verify the importance of long distance dependency relationships in the feature extraction process, we disable the use of TSS. The phased processing method loses its effectiveness in the single CNN architecture of SLDDNet-2.

SLDDNet-3 is designed to verify the contribution of the position information captured by ASE to the model performance.

To verify the optimization of network performance by deep supervision mechanism, SLDDNet-4 cancels the calculation of multiscale loss and only uses the final predicted result to calculate the loss.

DF focuses on the difference features while completing the fusion of detail information and semantic information. In SLDDNet-5, we disable the use of this module and use a concatenated way to complete feature fusion.

2) LEVIR-CD: On the LEVIR-CD dataset, we conduct experiments on five ablation models, and partial experimental results are shown in Fig. 5. As indicated by the blue boxes, all five ablation models have varying degrees of problems in handling details. Although SLDDNet-2 has similar results to SLDDNet in the processing of the first sample, it shows obvious edge judgment errors in the processing of the second sample. This indicates that establishing only short distance dependency relationships cannot effectively capture change features.

Table II presents the performance metrics of the five ablation models and SLDDNet on the LEVIR-CD dataset. While the five ablation models achieve higher Recall than SLDDNet, they perform lower than SLDDNet in the other six metrics. Recall primarily reflects the proportion of actual true values in the positive samples obtained by the model. The lower recall of SLDDNet suggests that there is still room for improvement in its robustness. Overall, the modules and processing mechanisms corresponding to the five ablation models have a positive impact on the network's performance, as supported by subjective analysis. This indicates that the final network architecture that we have chosen is reasonable and effective.

3) WHU-CD: Fig. 6 illustrates the processing results of various ablation models on the WHU-CD dataset. This dataset has a high resolution, allowing for a better display of detailed information, but it also introduces more noticeable environmental interference factors. In the processing of the first sample, environmental factors affect the processing results of SLDDNet-1, SLDDNet-3, SLDDNet-4, and SLDDNet-5. Although SLDDNet-2 effectively avoids the influence of environmental factors, there is still a discernible gap in terms of feature integrity compared to SLDDNet. This highlights the significance of the long-range dependency captured by the TSS in the FR process. In the processing of the second sample, all five ablation models exhibit varying degrees of edge judgment problems, indirectly emphasizing the effectiveness of the SLDDNet architecture.

As shown in Table III, although some indicators of the ablation models are higher than those of SLDDNet, overall, the results of the five ablation models are not as good as

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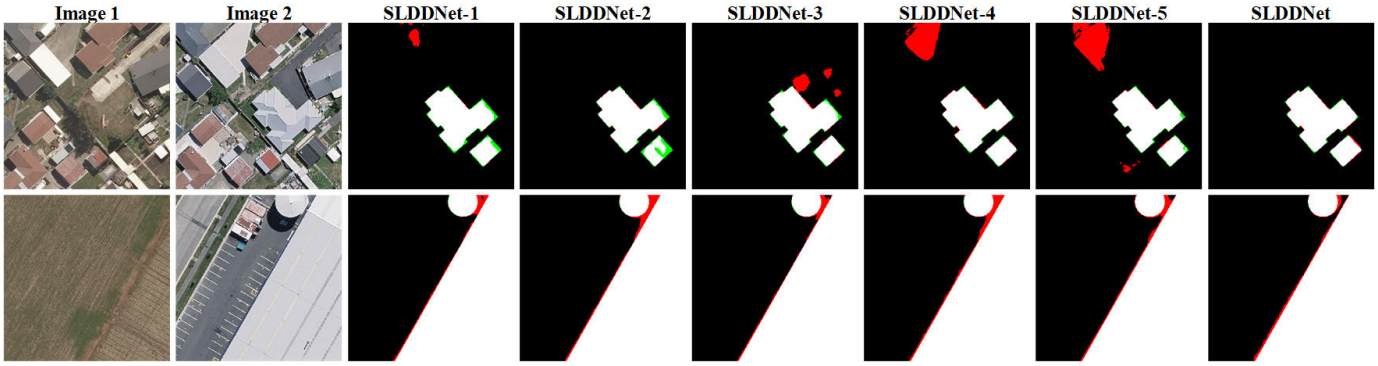


Fig. 6. Results of ablation experiments on WHU-CD.

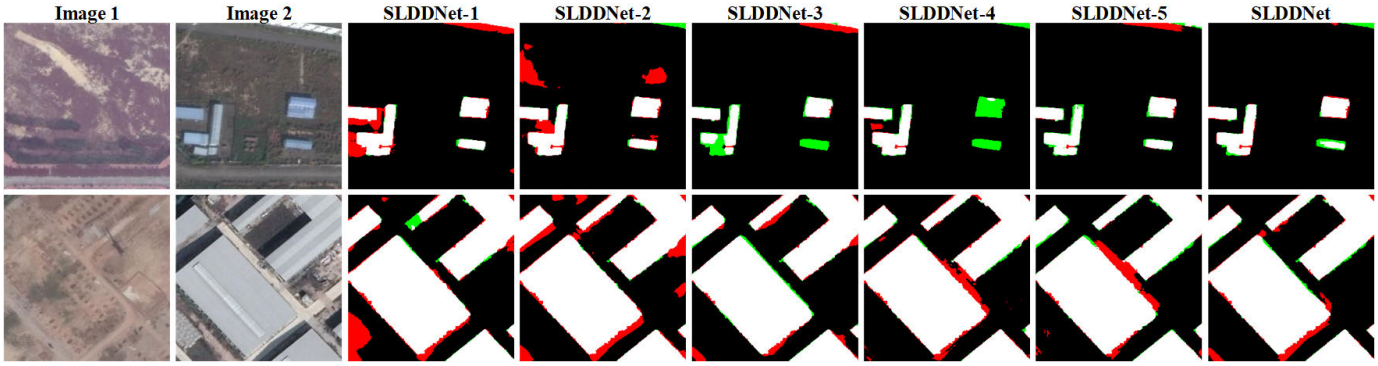


Fig. 7. Results of ablation experiments on GZ-CD.

TABLE III
INDICATORS FOR FIVE ABLATION MODELS ON THE WHU-CD DATASET

Method	OA	AA	Kappa	F1	IoU	Precision	Recall
SLDDNet-1	99.28	99.28	92.06	92.46	85.97	93.42	91.52
SLDDNet-2	99.22	99.22	91.29	91.70	84.67	93.87	89.62
SLDDNet-3	99.26	99.26	91.76	92.15	85.45	93.69	90.66
SLDDNet-4	99.22	99.22	90.75	91.89	84.99	92.41	91.37
SLDDNet-5	99.28	99.27	91.45	92.42	85.90	93.27	91.61
SLDDNet	99.26	99.27	92.11	92.76	85.78	92.27	92.43

TABLE IV
INDICATORS FOR FIVE ABLATION MODELS ON THE GZ-CD DATASET

Method	OA	AA	Kappa	F1	IoU	Precision	Recall
SLDDNet-1	97.38	97.38	84.21	85.65	74.90	90.12	81.61
SLDDNet-2	96.95	96.95	82.38	84.07	72.52	84.08	84.06
SLDDNet-3	97.47	97.47	84.97	86.53	76.26	88.24	84.88
SLDDNet-4	97.40	97.40	84.31	86.10	75.59	88.33	83.98
SLDDNet-5	97.52	97.52	84.88	86.40	76.06	90.95	82.29
SLDDNet	97.51	97.51	85.24	86.61	76.38	89.31	84.07

SLDDNet. Combined with the analysis of the results in Fig. 6, TSS effectively improves the network's ability to pay attention to global semantic information. ASE and DF make important contributions to the position information of the changing features.

4) GZ-CD: As shown in Fig. 7, the results of the final model and the five ablation models on the GZ-CD dataset are presented. In the first group of samples, SLDDNet-1 shows rough processing at the target edge, while, in the second group of samples, it misidentifies the object in the bottom left corner as a building. This demonstrates the important role of PSFS in capturing local features and reconstructing small targets. In contrast, SLDDNet-2 contains a large amount of environmental noise in its processing of the first group of samples. Although SLDDNet-2 performs better than SLDDNet-1 in processing the second group of samples, it still contains environmental noise, indicating that the long-range dependency relationship between the feature information captured by TSS can help the network better identify changing targets. In addition, SLDDNet-3, SLDDNet-4, and SLDDNet-5 all have varying degrees of edge determination and changing target recognition issues, which indicates that ASE, deep supervision mechanism, and DF all contribute to the performance of the model and also prove the rationality of the SLDDNet architecture.

As shown in Table IV, the comprehensive indicator results analysis shows that SLDDNet performs well compared to the five ablation models on the GZ-CD dataset. However, there is still some gap between the results obtained by SLDDNet and the GT, indicating that there is still room for improvement in SLDDNet's performance. In the future, we will continue to optimize based on SLDDNet to achieve high-quality processing.

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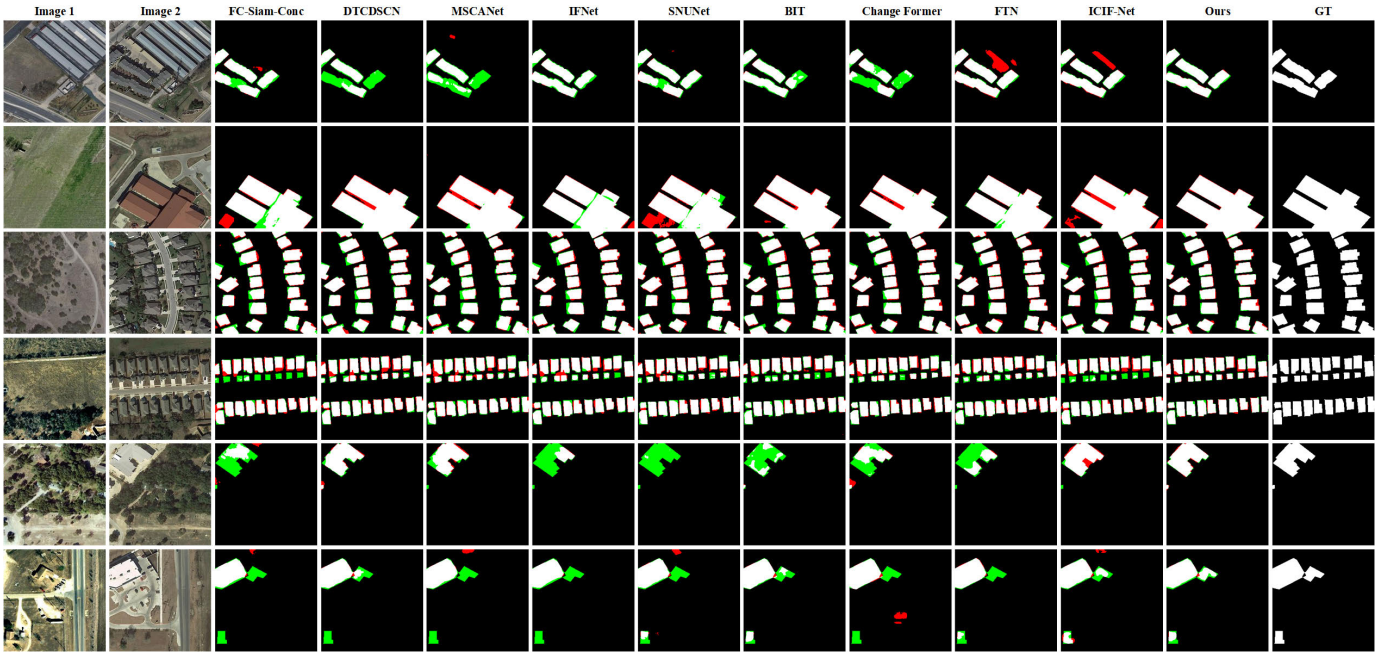


Fig. 8. Comparative experiments on the LEVIR-CD dataset.

This section of this article presents only the five most representative ablation models and experimental data. Through ablation experiments, we ultimately determined the network architecture, which is the SLDDNet.

E. Comparison Experiment

After determining the final network architecture, we compare SLDDNet with current mainstream and state-of-the-art methods on three datasets. The maximum number of training epochs for all participating comparison methods is 200, and the best model is updated at the end of each training epoch, while the learning rate is set to 0.001, which is commonly used in the field of CD. The other parameters of each comparison method are assigned values as described in their respective papers. We will describe the experiments grouped by datasets. All experimental results are visually overlaid with GT, where red represents misidentified areas and green represents unrecognized areas.

1) LEVIR-CD: Fig. 8 shows the results obtained by various methods on the LEVIR-CD dataset. The first and second groups of samples contain large change targets. DTCDSCN, MSCANet, change former, and other networks have obvious target disconnection problems in the first group of samples. Although FTN and ICIF-Net perform well in detecting change targets, it also outputs environmental interference factors as targets. In contrast, Ours achieves results closest to GT.

In the second group of samples, Ours shows excellent performance in edge processing and target continuity. BIT, change former, ICIF-Net, and other networks are inferior to Ours in edge processing. Due to the influence of lighting, shadows appear on the change targets. SNUNet, FTN, and other networks mistakenly identify shadows as boundaries, resulting in feature discontinuities in the final results.

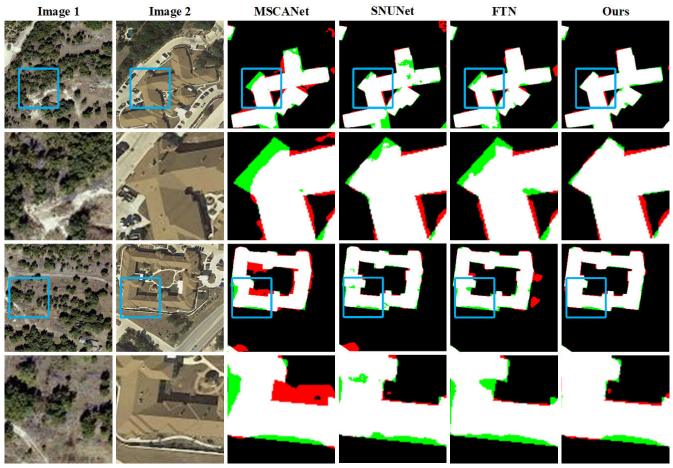


Fig. 9. Detailing of large change targets.

In the fourth group of samples, dense residential areas are included. From the processing results, IFNet, BIT, ICIF-Net, and other networks all lack the detection of small change targets. Ours not only performs better in edge processing than all the comparison methods but also performs well in small target detection.

Since the acquisition time of the bitemporal images is different, the targets in the sixth group of samples have different colors. Although there is a certain gap between the results obtained by Ours and GT in terms of edge processing, compared with networks such as change former and FTN that have missing targets, Ours is still the best among all comparison methods.

To highlight SLDDNet's ability to handle large-scale change targets, we additionally select two sets of samples for local amplification processing, as shown in Fig. 9. MSCANet, SNUNet, and FTN exhibit significantly less effectiveness in

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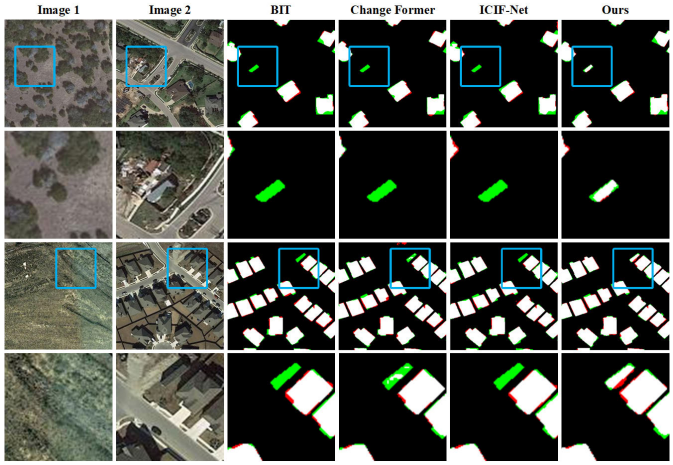


Fig. 10. Local enlargement for small target processing.

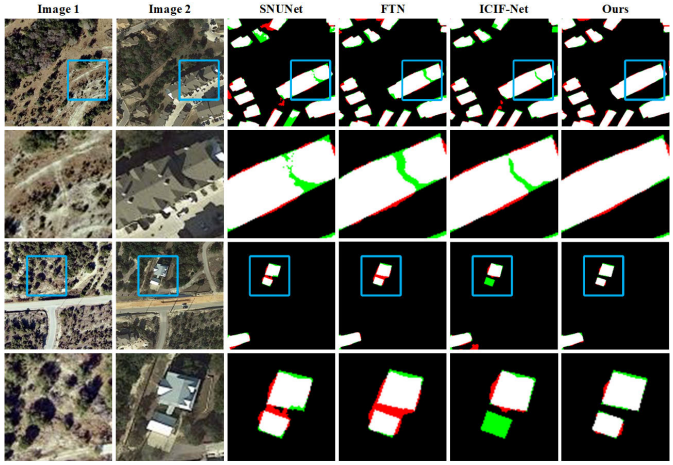


Fig. 11. Comparison of local details of the treatment of environmental factors.

determining the edges of targets compared to Ours primarily due to environmental factors and illumination. Although Ours achieves better results than the compared methods, there remains considerable room for improvement in edge processing.

Fig. 10 illustrates two sets of samples, highlighting the detection capability of SLDDNet for small-scale changed targets. BIT, change former, and ICIF-Net exhibit significant deficiencies in detecting small targets. In contrast, our SLDDNet accurately identifies small-scale changed targets, thanks to the establishment of both long and short-range dependencies and the stagewise exchange of information.

Regarding the data, shadows resulting from lighting and color variations are significant factors that impact the accuracy of remote sensing CD. To showcase SLDDNet's capability in handling shadows and color changes, we choose two representative samples for displaying local details, as shown in Fig. 11. In the first sample, a substantial shadow is formed in the middle of the building due to lighting conditions. SNUNet, FTN, and ICIF-Net exhibit noticeable segmentation errors in the change target due to the influence of shadows.

In contrast, Ours effectively captures long-range dependencies between feature information using TSS. When faced with large-scale changing targets, it can establish connections even among distant target features. PSFS focuses on addressing short-range dependencies between feature information through local processing, allowing Ours to demonstrate desirable performance in handling detailed information and small targets. The global semantic information captured by TSS and the local semantic information captured by PSFS are combined through staged aggregation during the feature extraction stage, resulting in a more comprehensive preservation of target feature information. As a result, it can better handle shadow issues in the first set of samples.

In the second set of samples, the three compared networks exhibit varying degrees of errors in edge detection and target recognition due to the influence of lighting conditions and the volumes of buildings. Due to the utilization of TSS, Ours demonstrates superior capability in capturing feature information of changing targets in complex environments. Furthermore, through the implementation of PSFS, Ours effectively enhances its ability to acquire local feature information, leading to improved semantic associations for small targets. As a result, Ours demonstrates excellent performance.

TABLE V
INDICATOR RESULTS FOR EACH COMPARISON METHOD ON THE LEVIR-CD DATASET

Method	OA	AA	Kappa	F1	IoU	Precision	Recall
FC-Siam-Conc [34]	98.69	98.69	86.56	87.25	77.39	86.37	88.15
DTCDSCN [42]	99.00	99.00	89.56	90.09	81.97	91.18	89.02
MSCANet [51]	98.86	98.86	88.06	88.67	79.64	89.79	87.57
IFNet [39]	98.95	98.95	88.77	89.32	80.70	92.57	86.29
SNUNet [64]	98.67	98.66	89.38	86.32	75.93	90.33	82.65
BIT [45]	99.03	99.03	89.83	90.34	82.39	91.53	89.19
Change Former [46]	99.04	99.04	89.89	90.40	82.48	92.05	88.81
FTN [54]	99.06	99.09	**90.63**	91.01	83.51	**92.71**	89.37
ICIF-Net [47]	**99.12**	**99.10**	89.55	**91.18**	**83.85**	91.13	**90.57**
SLDDNet	**99.18**	**99.16**	**91.27**	**91.75**	**84.76**	**93.64**	**89.94**

Red for best results, blue for second best results.

Table V displays the performance metrics of all the methods evaluated on the LEVIR-CD dataset. SLDDNet achieves the highest of the six metrics. Notably, SLDDNet outperforms FTN by 0.64 points in terms of kappa, showcasing its superior performance. In F1 score, SLDDNet surpasses ICIF-Net, the second-ranked method, by 0.57 points. Similarly, SLDDNet outperforms ICIF-Net by 0.91 points in IoU. Based on both subjective and objective evaluations, SLDDNet demonstrates excellent performance on the LEVIR-CD dataset. LDDNet surpasses mainstream and state-of-the-art methods in target recognition, edge processing, and small target detection. However, SLDDNet does not perform as well in handling some complex samples, indicating that there is still room for improvement in the robustness of SLDDNet.

2) WHU-CD: Fig. 12 showcases partial experimental results on the WHU-CD dataset. The first group of samples consists of two large change targets with distinct colors.

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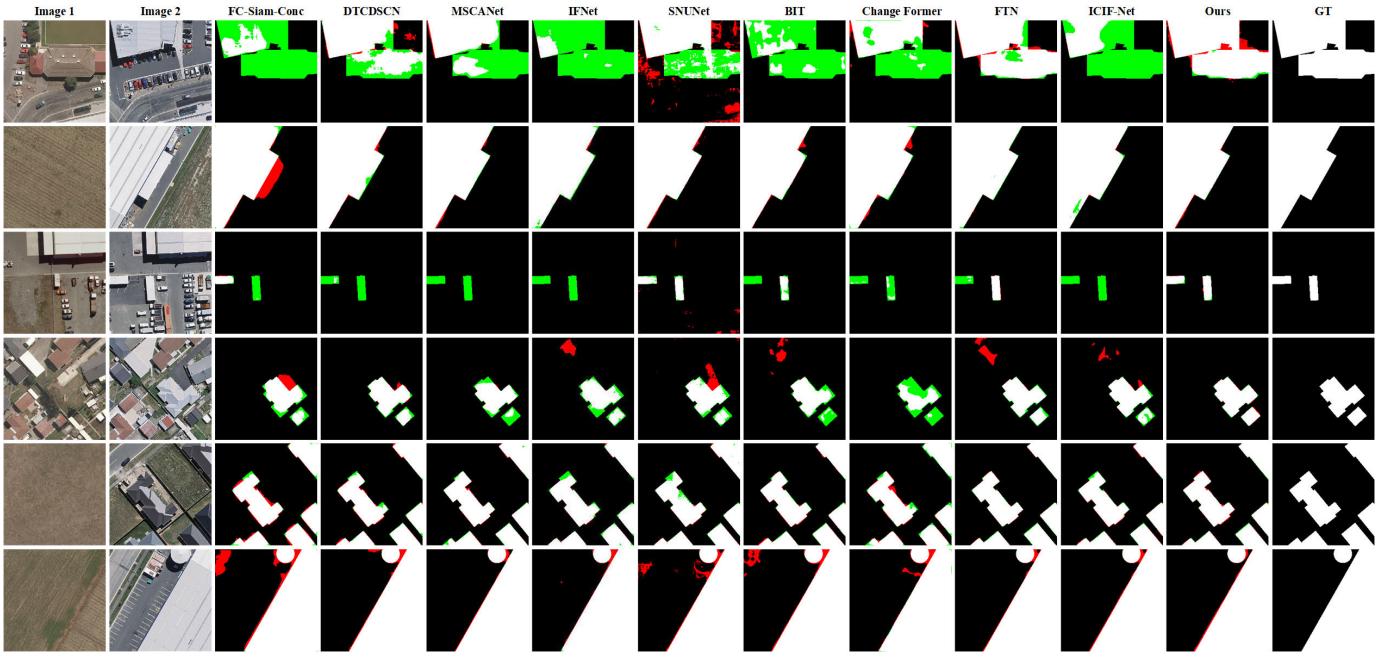


Fig. 12. Comparative experiments on the WHU-CD dataset.

Networks such as SNUNet, change former, and ICIF-Net successfully recognize buildings that exhibit noticeable color differences from their surroundings. However, when faced with a different type of building that shares a similar color with the environment, all three networks fail to maintain their recognition capability.

While Ours exhibits some errors in edge detection, it manages to complete the detection of two large buildings fairly comprehensively. This is mainly attributed to TSS assisting the network in modeling feature information from a global perspective during the feature extraction stage. It enables the effective establishment of correlations between distant features of large changing targets. In addition, PSFS assists the network in establishing associations between feature information from a local perspective, assigning different weights to the feature information of large targets and their surrounding environment. The staged aggregation of long- and short-range dependencies effectively integrates global semantic information and local semantic information, facilitating feature interaction and ultimately producing results that are closer to the GT.

The third group of samples is a parking lot with many vehicles, where the size and dense arrangement of the buildings are similar to that of the cars. This environmental factor poses a higher challenge to the network's performance. Networks such as change former, BIT, and ICIF-Net all have different degrees of target missing. Although SNUNet and BIT complete the detection of the changing targets, they suffer from noise both inside and around the buildings due to environmental factors.

Of particular importance is the feature extraction stage for judging and capturing changing targets. SNUNet and BIT, which only employ simple CNN architectures for feature extraction, are unable to model the relationships between feature information from a global perspective. As a result, these methods tend to have more noise in their results. Although change former and FTN capture global semantic information using self-attention in the feature extraction stage, the lack of capturing local feature information leads to errors in recognizing changing targets. In contrast, Ours adopts a staged aggregation approach to fuse the global semantic information captured by TSS and the local semantic information captured by PSFS. This enables the network to effectively recognize and capture changing targets, thereby achieving desirable performance in handling small changing targets.

The fourth group of samples contains densely arranged residential areas with diverse colors of buildings, which adds difficulty to the CD task. FC-Siam-Conc and change former visibly lack edge shape in their processing results. Due to the influence of the multicolor environment, networks such as SNUNet, BIT, FTN, and ICIF-Net currently struggle with edge determination and produce a significant amount of noise in the final result image.

To highlight the detection ability for large-scale change targets, we select two sets of samples for local detail comparison, as shown in Fig. 13. In the processing of the first group of samples, ICIF-Net is clearly influenced by the surrounding vehicles and misclassifies them as buildings. Although FTN's results in edge processing are close to those obtained by Ours in the first group of samples, FTN noticeably lags behind Ours in edge determination in the second group of samples. In the second group, Ours also exhibits irregular jagged edge determination but outperforms the networks utilizing the three Transformer architectures in terms of maintaining target integrity.

Ours' ability to exhibit desirable performance in detecting large-scale changing targets depends on the staged aggregation pattern proposed during the feature extraction stage. This approach combines global semantic information and local semantic information at the same scale, enabling more

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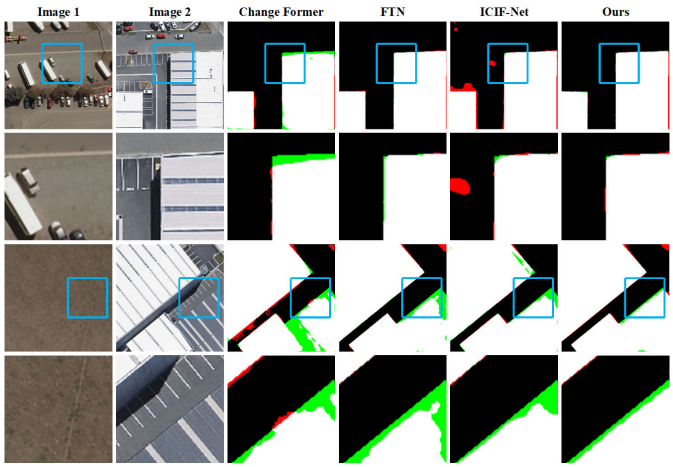


Fig. 13. Detailing of large change targets.

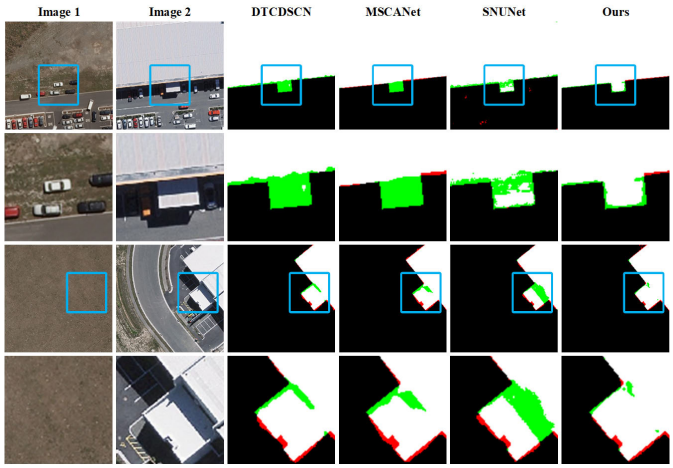


Fig. 14. Comparison of local details of the treatment of environmental factors.

effective interaction between local information and global information. Compared to the CNN-Transformer separate feature extraction approach used in current advanced methods in the CD field, the approach adopted by Ours has more prominent advantages.

To emphasize the handling of environmental factors, such as lighting, we select two representative samples for a local detail comparison, as shown in Fig. 14. The illuminated parts of the buildings in the two samples are clearer compared to the areas obscured by shadows caused by lighting. All four methods, including our proposed method, are affected to varying degrees. Overall, our method achieves a relatively comprehensive detection of buildings although there is room for improvement in edge processing. However, the other three comparison methods all show varying degrees of feature loss.

Furthermore, to emphasize the handling of dense buildings, we select two representative samples for local detail amplification, as depicted in Fig. 15. The buildings in both samples exhibit varied colors. In addition, due to the impact of lighting, the colors of different roof slopes also differ. SNUNet fails to detect certain areas of the buildings due to this environmental factor. Although change former and ICIF-Net can detect most buildings, their performance in edge processing is suboptimal.

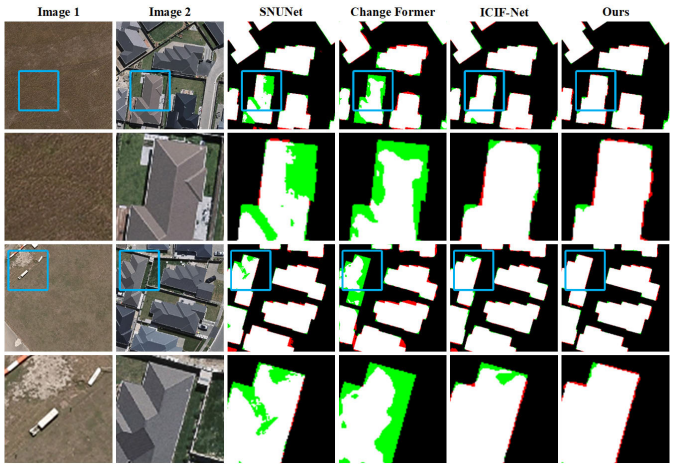


Fig. 15. Detail of a dense building in partial enlargement.

TABLE VI
INDICATOR RESULTS FOR EACH COMPARISON METHOD ON THE WHU-CD DATASET

Method	OA	AA	Kappa	F1	IoU	Precision	Recall
FC-Siam-Conc [34]	98.45	98.45	83.33	84.13	72.60	83.07	85.19
DTCDSCN [42]	99.05	99.05	89.53	90.03	81.87	91.46	88.55
MSCANet [51]	98.95	98.97	88.55	88.62	79.56	93.16	84.50
IFNet [39]	98.83	98.82	83.02	83.41	71.53	96.91	73.20
SNUNet [64]	98.91	98.91	87.77	88.34	79.11	91.34	85.53
BIT [45]	98.81	98.81	86.85	87.47	77.73	88.71	86.27
Change Former [46]	98.76	98.79	86.03	86.88	76.81	88.50	85.33
FTN [54]	**99.37**	**99.36**	**91.73**	**92.16**	**85.45**	**93.09**	**91.24**
ICIF-Net [47]	99.13	99.13	90.31	90.77	83.09	**92.93**	88.70
SLDDNet	**99.26**	**99.28**	**92.11**	**92.76**	**85.78**	92.27	**92.43**

Red for best results, blue for second best results.

According to Table VI, SLDDNet achieves the highest scores in all four metrics. Specifically, SLDDNet outperforms FTN by 0.6 points in F1 and 0.33 points in IoU. However, SLDDNet does not achieve optimal results on OA and precision, which indicates that there is still room for improvement. Overall, SLDDNet outperforms current mainstream and state-of-the-art networks on the WHU-CD dataset. Particularly, when confronted with challenging environmental factors, SLDDNet leverages the interactive nature of phase-dependent long and short-term dependencies to enhance feature information at each stage. In addition, the incorporation of the ASE module strengthens positional information within the features, thereby improving the localization of detection targets. However, there is still room for improvement in SLDDNet. Its handling of edge features in samples with complex environments, as seen in the first group of samples in Fig. 12, remains suboptimal. Addressing this limitation will be a focal point of our future endeavors.

3) GZ-CD: The experimental results on the GZ-CD dataset are in Fig. 16. In the processing of the first group of samples, BIT and change former are much less capable of recognizing

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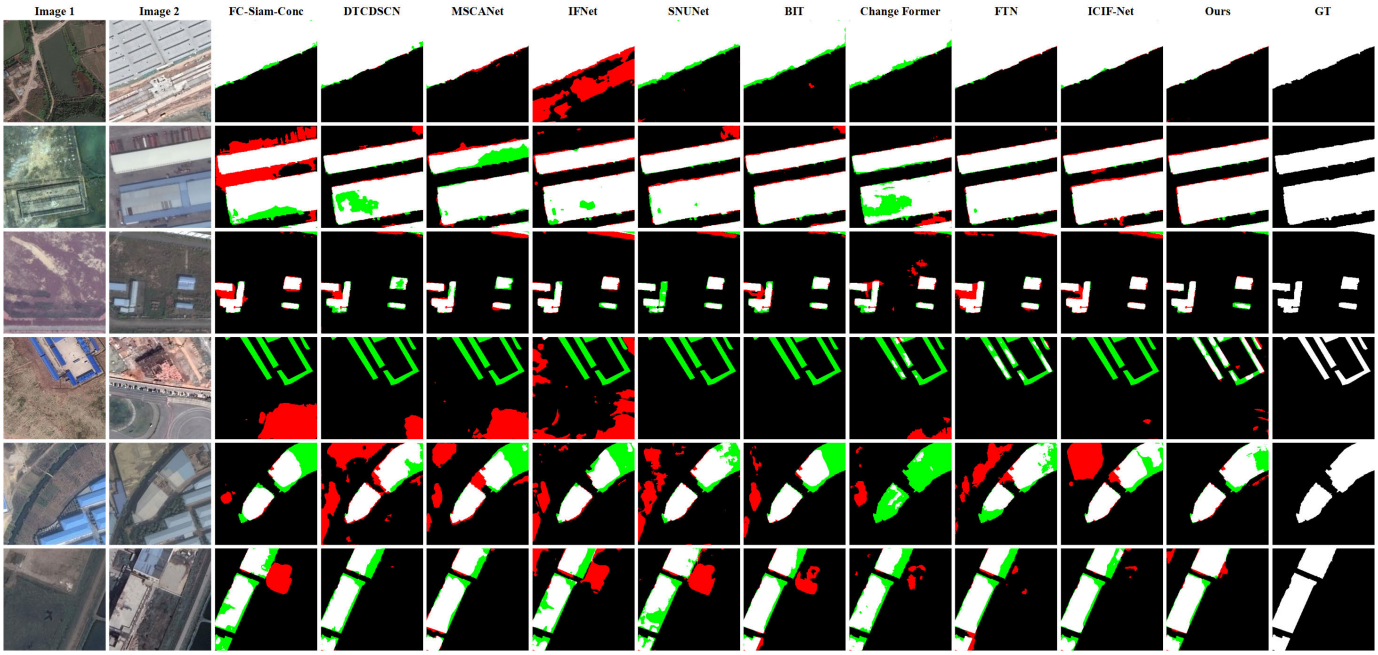


Fig. 16. Comparative experiments on the GZ-CD dataset.

the edges of changing targets than Ours, mainly due to the fact that both use a single CNN architecture or Transformer architecture to complete the feature extraction. Although FTN also uses only Swin Transformer as the backbone network to complete the feature extraction, Swin Transformer's sliding window, and multiscale processing enables better focus on feature information, FTN shows results close to Ours in the first group of samples.

TSS assists Ours in modeling feature information globally, while PSFS aids Ours in computing the correlation of feature information locally. When facing large-scale changing targets, such as those in the first group of samples, the staged aggregation network, which is built upon the parallel architecture of CNN and Transformer, can effectively integrate semantic information. This allows the features of large-scale changing targets to be assigned higher weights. In addition, the three-layer DF module achieves the fusion of deep semantic information and shallow detailed information, ultimately enabling Ours to obtain results that are closest to the GT.

The third and fourth groups of samples have fuzzy edges and relatively small change targets. BIT, change former, and ICIF-Net obtain results in the third group of samples that are significantly noisier and do not determine the edges of the targets well, while Ours obtains the closest results to GT. Although Ours does not perform well in terms of feature continuity in the fourth group of samples, it is the only network among the methods that are able to detect changing targets. In the sixth group of samples, networks such as IFNet, SNUNet, and BIT mistakenly identified the square open space in the environment as a building. Although there is also some noise in the results obtained by Ours, in terms of completeness, Ours is higher than all comparison methods. This further demonstrates that the architecture proposed by Ours is superior to the current mainstream and state-of-the-art approaches.

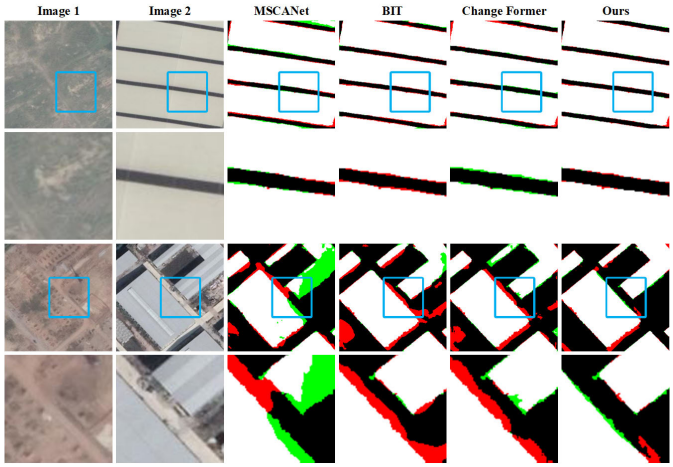


Fig. 17. Detailing of large change targets.

We use Fig. 17 to demonstrate SLDDNet's detailed processing of large-scale changed objects. In the treatment of the first group of samples, MSCANet, BIT, and change former show significantly more roughness on the edges. Due to the blurred edges of buildings in the second group of samples, all three comparison methods make incorrect edge judgments. However, the results obtained by Ours are closest to the GT.

As shown in Fig. 18, we select two sets of samples to highlight the ability to handle small targets. In the first group of samples, BIT and change former miss the detection of changing buildings, while ICIF-Net only identifies part of the building area. Ours, on the other hand, effectively processes local information and focuses on global semantic information by aggregating long and short distance dependencies during the feature extraction phase. This gives Ours an advantage over both mainstream and state-of-the-art methods in detecting

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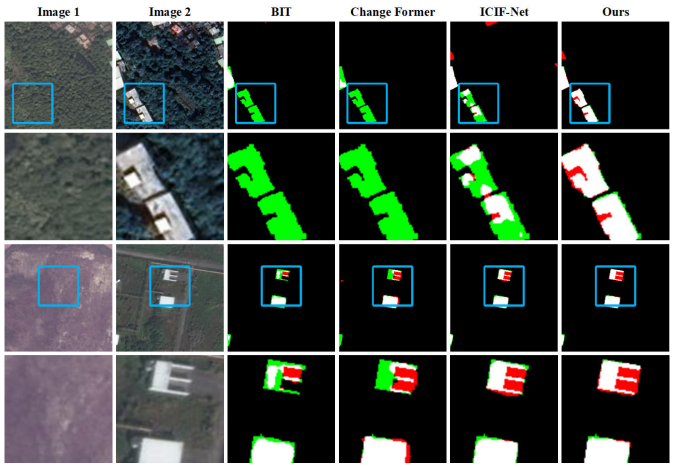


Fig. 18. Local enlargement for small target processing.

TABLE VII
INDICATOR RESULTS FOR EACH COMPARISON METHOD ON THE GZ-CD DATASET

Method	OA	AA	Kappa	F1	IoU	Precision	Recall
FC-Siam-Conc [34]	95.68	95.68	73.88	76.24	61.61	80.59	72.34
DTCDSCN [42]	96.83	96.83	80.98	82.72	70.53	86.65	79.13
MSCANet [51]	96.40	96.40	78.68	80.66	67.59	83.04	78.42
IFNet [39]	96.92	96.93	83.09	82.15	69.71	**92.19**	74.08
SNUNet [64]	97.07	97.07	82.64	84.25	72.79	86.82	81.82
BIT [45]	96.31	96.31	78.20	80.23	66.99	82.40	78.18
Change Former [46]	95.53	95.54	71.26	73.66	58.30	84.59	65.23
FTN [54]	**97.92**	**97.91**	82.14	**85.58**	**74.79**	86.99	**84.21**
ICIF-Net [47]	97.29	97.29	**83.60**	85.09	74.05	**89.90**	80.76
SLDDNet	**97.51**	**97.51**	**85.24**	**86.61**	**76.38**	89.31	**84.07**

Red for best results, blue for second best results.

small targets. Although Ours does not perform ideally in capturing the details of changing targets, it benefits from the introduction of ASE and achieves more accurate target localization. For example, in the second group of samples with complex and fine structures, all methods, including Ours, fail to obtain accurate results. However, Ours outperforms the three compared methods in terms of target completeness and target localization.

In Table VII, SLDDNet achieves top results in F1, kappa, and IoU, with second best results in OA, AA, and recall. Combining subjective and objective evaluations, SLDDNet outperforms mainstream and state-of-the-art methods on the GZ-CD dataset. However, it also reveals current problems of SLDDNet, such as detecting fuzzy edges (see Fig. 16, group 5) and handling complex objects (see Fig. 18, group 2). In the future, we continue to optimize the network's feature extraction and target positioning capabilities to address these issues.

Overall, we present a comparison experiment from both subjective and objective perspectives. Through experiments on three datasets, we demonstrate the excellent performance of SLDDNet in handling adversarial environmental interference and processing edge information, surpassing mainstream, and state-of-the-art methods. However, we also identify shortcomings of SLDDNet, such as handling blurred edges and complex small targets. In the future, we focus on improving the ability to handle fuzzy features.

4) Complexity and Parameter Number: In Table VIII, SLDDNet outperforms all the attention-based methods in terms of the number of parameters and floating-point computations. Although SLDDNet has a higher number of parameters and floating-point computations compared to the CNN-based architecture FC-Siam-Conc, its performance achieved is much higher than FC-Siam-Conc. Compared to the hybrid CNN-Transformer architecture, SLDDNet achieves the best results in terms of floating-point computations and the second-best results in terms of the number of parameters. This indicates that there is still room for optimization in SLDDNet.

Furthermore, Table VIII also records the time taken by each compared method to complete the processing of a set of samples with a size of 256 × 256 using a single GPU. Although SLDDNet has a slight advantage in terms of time complexity compared to FTN, it still has room for further improvement compared to methods such as BIT and ICIF-Net. In the future, we will continue to optimize the network structure, reduce computational effort, and decrease the time complexity of SLDDNet while maintaining its performance.

V. DISCUSSION

This article proposes an SLDDNet for remote sensing CD. SLDDNet proposes two modules, PSFS and TSS, to capture short and long distance dependencies, respectively, enhancing the network's ability to recognize both small- and large-scale changes in targets. The staged processing in SLDDNet allows the fusion of features captured by PSFS and TSS, which contains different semantic information, enabling better information interaction. This addresses the limitation of current state-of-the-art methods that struggle to effectively integrate feature information in hybrid CNN and Transformer architectures. To focus on change targets, SLDDNet proposes the DF that leverages detailed information from shallow networks to complement the semantic information from deep networks, improving the reconstruction of change targets. ASE is introduced to enhance position information, while DO-Conv is used instead of traditional convolution to achieve better training results. The deep supervision mechanism in SLDDNet allows for loss calculation from multiple scale outputs, optimizing the training process.

In addition, to validate the effectiveness of SLDDNet and its ability to handle CD tasks, we discuss the processing results in real-world scenarios. We use Google Earth to obtain images of the Oxford area in Mississippi, USA, from 2012 to 2019. In Fig. 19, we present a comparison between Ours and the latest methods. BIT's edge prediction for buildings is impacted by lighting conditions. The results obtained by change former and ICIF-Net miss certain areas of the buildings. While FTN's results are superior to the other three CNN-Transformer hybrid

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TABLE VIII
COMPLEXITY AND PARAMETER NUMBER

Method	Complexity			LEVIR-CD		WHU-CD		GZ-CD	
	Params(M)	FLOPs(G)	Time(Second)	F1	IoU	F1	IoU	F1	IoU
FC-Siam-Conc [34]	1.55	5.32	0.02	87.25	77.39	84.13	72.60	76.24	61.61
DTCDSCN [42]	41.07	13.21	0.06	90.09	81.97	90.03	81.87	82.72	70.53
MSCANet [51]	16.59	14.70	0.04	88.67	79.64	88.62	79.56	80.66	67.59
IFNet [39]	50.71	82.35	0.05	89.32	80.70	83.41	71.53	82.15	69.71
SNUNet [64]	12.03	54.88	0.04	86.32	75.93	88.34	79.11	84.25	72.79
BIT [45]	3.55	10.59	0.05	90.34	82.39	87.47	77.73	80.23	66.99
Change Former [46]	41.03	202.87	0.13	90.40	82.48	86.88	76.81	73.66	58.30
FTN [54]	168.53	45.00	0.27	91.01	83.51	92.16	85.45	85.58	74.79
ICIF-Net [47]	25.83	25.27	0.12	91.18	83.85	90.77	83.09	85.09	74.05
SLDDNet	9.52	4.69	0.20	91.75	84.76	92.76	85.78	86.61	76.38

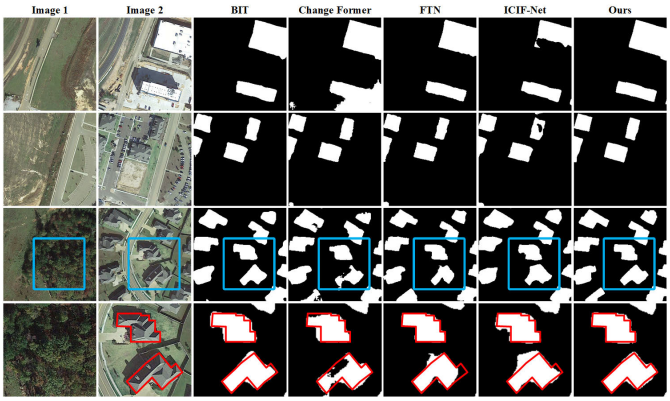


Fig. 19. Comparison chart of real scene image verification.

architecture networks, it still falls short in terms of edge processing compared to Ours. In Fig. 19, we zoom in on the third group of samples to focus on local details, where the outlines of buildings are manually labeled. Compared to the four latest methods, Ours demonstrates the best performance in edge discrimination and feature continuity.

VI. CONCLUSION

Although we demonstrate the good results of SLDDNet on three datasets in this article, there is still a certain gap between SLDDNet and GT. SLDDNet still has performance fluctuations when dealing with certain samples with blurred edges, which also indicates that there is still room for improving its performance. In addition, SLDDNet does not achieve the best results on some metrics in the comparison and ablation experiments. In the experiments on the three datasets, SLDDNet obtains lower recall on LEVIR-CD. This is mainly due to the fact that the SLDDNet network does not have a significant advantage in its own robustness and, on the other hand, due to the fact that LEVIR-CD has a high sample complexity and requires high performance of the model.

Considering that SLDDNet still has room for improvement in performance, we will enhance its learning of blurry and weak feature information in the future. We will further explore network architectures and optimize network operation efficiency to enhance the robustness of the network. Specifically, for the feature extraction of bitemporal images, we will attempt to utilize the idea of graph convolution to improve the network's attention to semantic information. In addition, we will also test the model's performance on other optical datasets and synthetic aperture radar datasets.

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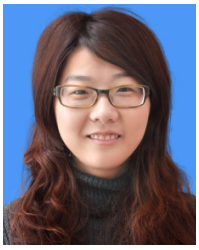
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